Static and Dynamic Evidence Supplement

Supplement to the post-results reconstruction manuscript

Final evidence supplement — primary/P0/P1 locked; reviewer controls complete

1 Scope and provenance

Tables S1–S18 retain the accepted static, primary, and P0/P1 evidence. Tables S19 onward add the prospectively frozen reviewer controls R1/R2 and post-hoc R4 diagnostic. R3 is retained as a predeclared protocol stop, not silently omitted. No classifier or frozen policy was retrained, and no primary result was overwritten.

The static-stage original large archive was not re-audited in this revision. Its retained independent-review artifacts remain the numerical source for Tables S1–S7; the current adaptive archive supplies the later sections. These evidence layers must not be conflated. The six-file executed-source reconciliation has since been closed by an archival-only V2 recovery package: all six historical source files match the runtime-bound SHA-256 values exactly, and the later source copies remain separately retained. The post-hoc P1 strengthening archive is now complete and is incorporated in the final sections below. The primary V1 campaign remains unchanged and retains its own evidence identity.

2 Dataset grouping and evaluated coverage

The retained split design differs by dataset. TON-IoT groups records by unordered IP pair; this is not a host-disjoint or chronological split. CICIoT2023 separates capture files within the grouping policy; categories with insufficient captures remain training-only. N-BaIoT separates physical-device identifiers into five training, two validation, and two test devices. Saved source-row identifiers and raw-field-string fingerprints are disjoint across partitions; this does not imply that all numeric encoded vectors are distinct.

CICIoT2023 has 34 original labels in the available classification taxonomy, comprising 33 attacks and normal traffic. The actual test coverage is **17 labels: 16 attack categories plus Benign_Final**. Seventeen other attack categories are training-only. This is a binary attack-versus-normal evaluation, not a 34-class classification result.

Table S1. CICIoT2023 evaluated and training-only original labels. Columns are independent lists; rows do not denote matched categories.

Test labels	Training-only attack labels
Benign_Final	Backdoor_Malware
DDoS-ACK_Fragmentation	BrowserHijacking
DDoS-ICMP_Flood	CommandInjection
DDoS-ICMP_Fragmentation	DDoS-HTTP_Flood
DDoS-PSHACK_FLOOD	DDoS-SlowLoris
DDoS-RSTFINFLOOD	DNS_Spoofing
DDoS-SYN_Flood	DictionaryBruteForce
DDoS-SynonymousIP_Flood	DoS-HTTP_Flood
DDoS-TCP_Flood	MITM-ArpSpoofing
DDoS-UDP_Flood	Recon-HostDiscovery
DDoS-UDP_Fragmentation	Recon-OSScan
DoS-SYN_Flood	Recon-PingSweep
DoS-TCP_Flood	Recon-PortScan
DoS-UDP_Flood	SqlInjection
Mirai-greeth_flood	Uploading_Attack
Mirai-greip_flood	VulnerabilityScan
Mirai-udpplain	XSS

3 Device partition and encoded-vector diagnostic

Table S2. N-BaIoT device allocation in the frozen partition. These are dataset device names, not additional physical test platforms in this study.

Partition	Dataset device identifier
Train	Ecobee_Thermostat
Train	Ennio_Doorbell
Train	Philips_B120N10_Baby_Monitor
Train	Provision_PT_737E_Security_Camera
Train	SimpleHome_XCS7_1003_WHT_Security_Camera
Validation	Danmini_Doorbell
Validation	Samsung_SNH_1011_N_Webcam
Test	Provision_PT_838_Security_Camera
Test	SimpleHome_XCS7_1002_WHT_Security_Camera

A post-hoc byte-level audit of saved float32 inputs identified 0 TON-IoT, 13 CICIoT2023, and 14,449 N-BaIoT test rows with an exact training-vector match. The N-BaIoT matches constitute 17.3343% of its test rows and seven distinct encoded vectors. A further 21,832 N-BaIoT validation rows match training vectors. Distinct source-row identifiers and raw-field fingerprints can map to the same numeric input, including after preprocessing and float32 conversion. The matching vectors alone do not establish reuse of the same source row or the same physical device across partitions.

Table S3 excludes only test vectors matching *training* inputs. Models, scores, and thresholds remain unchanged. The supplied CSV additionally retains the separate training-or-validation-match diagnostic, which is not substituted for Table S3 or the main result. All excluded N-BaIoT rows are attacks; its benign confusion counts and FPR are consequently unchanged. F1 changes also reflect the changed class mixture. Diagnostic stability does not eliminate all possible dataset bias.

Table S3. Post-hoc sensitivity excluding exact training-vector matches. Primary F1 uses the original test set; diagnostic F1, BA, and FPR use retained rows. BA denotes balanced accuracy; FPR is a percentage.

Dataset	Model	Retained	Primary F1	Diagnostic F1	BA	FPR (%)
TON-IoT	TinyDT	22,490	0.479233	0.479233	0.659889	67.780
	LightLR	22,490	0.961952	0.961952	0.976626	1.354
	MedRF	22,490	0.974551	0.974551	0.990638	1.518
	HeavyMLP	22,490	0.973593	0.973593	0.988261	1.378
CICIoT2023	TinyDT	71,132	0.990371	0.990369	0.829023	34.195
	LightLR	71,132	0.987987	0.987985	0.786304	42.738
	MedRF	71,132	0.991545	0.991544	0.850052	29.990
	HeavyMLP	71,132	0.993015	0.993014	0.876306	24.739
N-BaIoT	TinyDT	68,906	0.998373	0.997985	0.986099	2.764
	LightLR	68,906	0.998393	0.998010	0.986632	2.646
	MedRF	68,906	0.999733	0.999669	0.998095	0.366
	HeavyMLP	68,906	0.997761	0.997228	0.980565	3.874

The original held-out results remain the primary results. This analysis was conducted after inspecting the encoded inputs and must not be described as a preregistered alternative split, a model-selection procedure, or proof of complete input deduplication.

4 Static replay: corrected interval convention

Each table entry summarizes ten replay seeds conditional on a fixed classifier, fixed preprocessing, and fixed held-out pool. The interval is

$$\bar{x} \pm t_{0.975,9}\frac{s}{\sqrt{10}}, \qquad t_{0.975,9} = 2.2621571627.$$

These are not ten independent classifier-training replications or ten physical experiments. The CSV retains unrounded values and all three corrected metric families: F1, balanced accuracy, and AUROC. The original mixture of normal-approximation output and Student-t analysis is not carried into these tables.

Table S4. All twenty static replay F1 aggregates. The source scenario identifier `drift` is reported as *prevalence-shift*, reflecting the actual manipulation rather than implying general concept-drift adaptation.

Dataset	Replay	Model	Mean F1	SD	Student-t 95% CI
CICIoT2023	Prevalence-shift	HeavyMLP	0.8090	0.0019	[0.8077, 0.8104]
CICIoT2023	Prevalence-shift	LightLR	0.7104	0.0018	[0.7091, 0.7117]
CICIoT2023	Prevalence-shift	MedRF	0.7778	0.0010	[0.7770, 0.7785]
CICIoT2023	Prevalence-shift	TinyDT	0.7542	0.0010	[0.7535, 0.7549]
CICIoT2023	Stationary	HeavyMLP	0.9931	0.0004	[0.9928, 0.9934]
CICIoT2023	Stationary	LightLR	0.9881	0.0007	[0.9876, 0.9887]
CICIoT2023	Stationary	MedRF	0.9917	0.0006	[0.9913, 0.9921]
CICIoT2023	Stationary	TinyDT	0.9904	0.0007	[0.9899, 0.9909]
N-BaIoT	Stationary	HeavyMLP	0.9977	0.0001	[0.9976, 0.9978]
N-BaIoT	Stationary	LightLR	0.9984	0.0001	[0.9983, 0.9984]
N-BaIoT	Stationary	MedRF	0.9998	0.0001	[0.9997, 0.9998]
N-BaIoT	Stationary	TinyDT	0.9984	0.0002	[0.9983, 0.9985]
TON-IoT	Prevalence-shift	HeavyMLP	0.9822	0.0008	[0.9816, 0.9828]
TON-IoT	Prevalence-shift	LightLR	0.9704	0.0007	[0.9699, 0.9710]
TON-IoT	Prevalence-shift	MedRF	0.9839	0.0006	[0.9834, 0.9843]
TON-IoT	Prevalence-shift	TinyDT	0.6068	0.0010	[0.6060, 0.6075]
TON-IoT	Stationary	HeavyMLP	0.9733	0.0004	[0.9730, 0.9737]
TON-IoT	Stationary	LightLR	0.9616	0.0005	[0.9612, 0.9619]
TON-IoT	Stationary	MedRF	0.9744	0.0006	[0.9740, 0.9748]
TON-IoT	Stationary	TinyDT	0.4791	0.0010	[0.4784, 0.4798]

Stationary traces contain 200 windows of 100 rows. Prevalence-shift traces contain 500 windows: successive phase lengths are 100, 100, 120, 100, and 80 windows, with attack proportions 0.05, 0.35, 0.70, 0.40, and 0.10. The replay seeds are 11, 23, 37, 53, 71, 89, 107, 131, 157, and 191. Prevalence-shift traces exist for TON-IoT and CICIoT2023, not for N-BaIoT in this evidence package. Each trace is evaluated with each fixed classifier, giving 200 completed static model–trace combinations.

The class-mixture schedule is not an online policy input. None of the entries in Table S4 establishes that a scheduler detected a phase change or adapted to an unknown attack. No adaptive policy was evaluated in this static stage.

5 Static replay: balanced accuracy

Table S5. Balanced accuracy for the same twenty static replay aggregates. Intervals describe replay resampling under fixed trained artifacts.

Dataset	Replay	Model	Mean BA	SD	Student-t 95% CI
CICIoT2023	Prevalence-shift	HeavyMLP	0.8762	0.0015	[0.8752, 0.8773]
CICIoT2023	Prevalence-shift	LightLR	0.7862	0.0019	[0.7849, 0.7875]
CICIoT2023	Prevalence-shift	MedRF	0.8502	0.0009	[0.8495, 0.8508]
CICIoT2023	Prevalence-shift	TinyDT	0.8291	0.0009	[0.8284, 0.8297]
CICIoT2023	Stationary	HeavyMLP	0.8773	0.0048	[0.8739, 0.8807]
CICIoT2023	Stationary	LightLR	0.7870	0.0069	[0.7820, 0.7920]
CICIoT2023	Stationary	MedRF	0.8518	0.0060	[0.8474, 0.8561]
CICIoT2023	Stationary	TinyDT	0.8286	0.0058	[0.8244, 0.8328]
N-BaIoT	Stationary	HeavyMLP	0.9800	0.0011	[0.9792, 0.9808]
N-BaIoT	Stationary	LightLR	0.9862	0.0008	[0.9856, 0.9868]
N-BaIoT	Stationary	MedRF	0.9984	0.0005	[0.9980, 0.9987]
N-BaIoT	Stationary	TinyDT	0.9867	0.0012	[0.9858, 0.9876]
TON-IoT	Prevalence-shift	HeavyMLP	0.9883	0.0006	[0.9879, 0.9888]
TON-IoT	Prevalence-shift	LightLR	0.9767	0.0007	[0.9762, 0.9772]
TON-IoT	Prevalence-shift	MedRF	0.9906	0.0004	[0.9903, 0.9909]
TON-IoT	Prevalence-shift	TinyDT	0.6604	0.0015	[0.6594, 0.6614]
TON-IoT	Stationary	HeavyMLP	0.9882	0.0003	[0.9880, 0.9884]
TON-IoT	Stationary	LightLR	0.9765	0.0004	[0.9762, 0.9768]
TON-IoT	Stationary	MedRF	0.9906	0.0003	[0.9904, 0.9908]
TON-IoT	Stationary	TinyDT	0.6598	0.0005	[0.6594, 0.6602]

The number of rows or windows is not an independent-replication count. These replay traces reuse a held-out pool and fixed predictions. Their confidence intervals do not incorporate training-seed variation, a different split, unseen devices, measurement calibration error, or a new environment. A bounded metric may yield a conventional unbounded Student-t endpoint; such an interval is not silently clipped or replaced by a different convention. Full precision is retained in the data file.

6 Uncertainty sources kept separate

Classifier performance is conditional on training seed 11. The ten replay seeds describe resampling and phase composition conditional on the saved model and pool. Static physical power variability is based on three accepted repetitions per profile. The completed dynamic comparison has ten paired blocks, with four methods within each block; it does not provide forty independent paired differences for one method comparison. Training variability, replay variability, and physical repeatability must remain separately named.

An interval crossing zero does not prove equivalence. A statement that detection loss is “small” requires an explicitly justified margin and an honest account of when it was selected. A margin chosen after inspecting dynamic test outcomes must not be presented as prespecified.

7 Pi prediction agreement and tight-loop timing

Table S6. Pi batch-inference timing. Each timing sample processes 100 already encoded rows; the tight loop follows twenty warm-up batches. The final column reports archived label disagreements in a separate 10,000-row parity check for each configuration, not a timing-sample count.

Dataset	Model	Timing batches	Median (ms)	p95 (ms)	Label differences
TON-IoT	TinyDT	1000	0.2448	0.3042	0
	LightLR	1000	0.3306	0.3977	0
	MedRF	1000	26.2880	27.4078	0
	HeavyMLP	1000	1.5923	1.6725	0
CICIoT2023	TinyDT	1000	0.2323	0.2889	0
	LightLR	1000	0.2999	0.3578	0
	MedRF	1000	26.0360	27.8218	0
	HeavyMLP	1000	1.5407	1.6281	0
N-BaIoT	TinyDT	1000	0.2557	0.3142	0
	LightLR	1000	0.3612	0.4275	0
	MedRF	1000	26.2434	27.3492	0
	HeavyMLP	1000	1.6570	1.7392	0

The timing boundary excludes network capture, feature extraction, preprocessing, and the scheduling policy. These batch times must not be relabeled as single-record or end-to-end IDS latency. The power study used one batch per second; that pacing differs from the tight-loop timing study. The archived parity reports are not new hardware tests executed for this writing draft.

8 Executed controller information boundary

The frozen controller state consists of smoothed previous temperature, the previously selected detector's mean attack probability, lagged queue utilization, previous arrival count scaled by 100, and previous action. Initial temperature is read from the device; initial threat context is 0.5, initial load is 1, and the initialized action is TinyDT. The four numerical fields are discretized into three bins each and combined with four previous actions, giving 324 possible states. Main-text methods state the exact boundaries and reward.

Test labels and trace phase identities are not policy inputs. Current-window predictions are retrieved or computed only for the selected action. Independent reconstruction reproduces every formal action from the recorded decision-available state using the frozen Q table, DQN weights, or CFSM rule. A label-informed single-step reference is separately identified as non-deployable; it is not a global trajectory oracle.

9 Recorded grouping function

The recorded TON-IoT rule hashes `11|` followed by the sorted IP addresses joined with `|`; the integer value of the first sixteen SHA-256 hexadecimal digits modulo 100 assigns values below 60 to training, 60–79 to validation, and 80–99 to test. CICIoT2023 sorts capture paths by SHA-256 of `11|` plus relative path and allocates approximately 60/20/20, with at least one capture per partition when three or more exist. N-BaIoT similarly hashes `11|` plus device name and assigns the first five, next two, and final two devices. Exact retained memberships, rather than a reconstructed rounding rule, bind the evaluation.

10 Accepted static physical measurements

Table S7. Fifteen accepted static segments. Duration and energy refer to the observed meter interval. Maximum temperature uses workload-window events only. Version identifies the retained execution/logging version, not a different classifier training.

Repeat	Profile	Version	Duration (s)	Energy (J)	Mean (W)	Max. (°C)
1	idle	v1	1800.001	4220.601	2.344777	33.102
1	TinyDT	v1	1799.997	4228.854	2.349367	33.102
1	HeavyMLP	v1	1800.004	4257.806	2.365442	34.076
1	LightLR	v1	1799.995	4224.380	2.346884	33.102
1	MedRF	v1	1800.002	4306.796	2.392662	33.589
2	idle	v1	1799.994	4194.960	2.330542	33.589
2	MedRF	v1	1799.996	4317.744	2.398752	33.102
2	LightLR	v1	1800.006	4226.906	2.348273	33.589
2	HeavyMLP	v1	1799.998	4248.284	2.360160	33.102
2	TinyDT	v2	1800.005	4244.018	2.357781	34.563
3	idle	v2	1799.995	4195.924	2.331076	33.102
3	LightLR	v3	1800.001	4223.704	2.346501	34.076
3	MedRF	v3	1799.998	4319.388	2.399662	35.050
3	HeavyMLP	v3	1800.000	4274.027	2.374460	35.537
3	TinyDT	v3	1800.000	4234.047	2.352248	36.024

All segments use one Pi 4B 8 GB and TON-IoT classifiers, with four models resident. The idle profile retains the instrumented process and model residency. Each active profile replays 10,000 unique training rows eighteen times in 1,800 batches of 100 rows at one batch per second. Fifteen complete segments give 27,000 workload windows and 27,015 boundary-inclusive meter observations. Two interrupted attempts remain excluded and retained separately rather than spliced into complete runs.

Energy is obtained from observed power and actual adjacent timestamps using trapezoidal integration. Raw current signs are preserved in the source record; the recorded orientation is interpreted consistently when computing positive consumed power. No new static-stage raw-frame reintegration was performed: the saved independent-review segment calculations are the numerical source. Dynamic-stage raw frames, by contrast, were independently decoded and integrated for this revision. Repetition-level means, standard deviations, and matched-idle differences are recomputed for the draft checks.

11 Measurement and archive boundaries

There is no independent laboratory calibration of the KM003C or independent ambient-temperature logger. Shared instrumentation and randomized paired ordering can improve comparability, but do not remove calibration uncertainty, environmental drift, carryover, or order effects. The low observed temperature range does not validate high-temperature operation or hard thermal guarantees. Whole-device USB-input power is not wall-plug AC energy, host energy, or a measured model-switching cost.

The historical root manifest has 371 valid listed entries. Two nested manifests were omitted:

```
raw/materialization/MANIFEST_SHA256.txt
raw/materialization/materialization_run_v2/MANIFEST_SHA256.txt
```

A complete historical content list contains 373 files excluding the root manifest itself. The retained `CONTENT_SHA256_373.txt` refers to that original archive, not this smaller manuscript package. This draft's own manifest is separate. The original archive is neither modified nor silently repackaged here.

12 Forty reconstructed physical runs

Table S8 retains every formal attempt reported in the campaign. All forty contain 500 windows, 501 power observations, and 50,000 predicted rows. Actual integration durations are retained at full precision in `data/dynamic/run_registry_recomputed.csv`; they are approximately 500 seconds rather than replaced by that nominal constant. No failed formal run appears in the supplied execution log. The 120-second pilot is separate.

Table S8. Actual within-block order and reconstructed run-level outcomes. Mean batch latency is inference of 100 encoded rows in milliseconds; it excludes isolated controller decision overhead.

Block	Order	Method	Energy (J)	F1	TN	FP	FN	TP	Batch (ms)
1	1	Tabular-Q	1192.491	0.969568	32342	458	585	16615	1.500
1	2	DQN	1200.351	0.969568	32342	458	585	16615	1.450
1	3	Static-MedRF	1210.268	0.983185	32282	518	68	17132	32.234
1	4	CFSM	1193.732	0.636830	13435	19365	118	17082	2.345
2	1	DQN	1197.694	0.971259	32336	464	523	16677	1.537
2	2	Static-MedRF	1207.126	0.983275	32280	520	63	17137	32.637
2	3	Tabular-Q	1191.666	0.971259	32336	464	523	16677	1.430
2	4	CFSM	1196.817	0.634207	13239	19561	130	17070	2.398
3	1	Tabular-Q	1190.721	0.969674	32350	450	589	16611	1.422
3	2	Static-MedRF	1205.739	0.983306	32278	522	60	17140	32.618
3	3	CFSM	1193.025	0.635446	13286	19514	103	17097	2.425
3	4	DQN	1200.791	0.969674	32350	450	589	16611	1.464
4	1	Tabular-Q	1190.126	0.969835	32361	439	594	16606	1.434
4	2	DQN	1194.741	0.969835	32361	439	594	16606	1.451
4	3	Static-MedRF	1209.233	0.984077	32295	505	50	17150	32.214
4	4	CFSM	1195.999	0.634658	13245	19555	115	17085	2.361
5	1	Static-MedRF	1206.460	0.983765	32285	515	51	17149	31.792
5	2	CFSM	1197.197	0.637593	13465	19335	102	17098	2.284
5	3	DQN	1201.542	0.969472	32345	455	591	16609	1.468
5	4	Tabular-Q	1194.920	0.969472	32345	455	591	16609	1.494
6	1	Tabular-Q	1191.543	0.971095	32398	402	587	16613	1.412
6	2	CFSM	1199.311	0.639080	13626	19174	119	17081	2.316
6	3	DQN	1207.523	0.971095	32398	402	587	16613	1.506
6	4	Static-MedRF	1203.668	0.985005	32333	467	55	17145	32.358
7	1	CFSM	1201.163	0.638346	13524	19276	100	17100	2.341
7	2	Tabular-Q	1194.960	0.970849	32348	452	548	16652	1.427
7	3	DQN	1195.812	0.970849	32348	452	548	16652	1.538
7	4	Static-MedRF	1211.791	0.983611	32294	506	65	17135	33.472
8	1	CFSM	1197.055	0.635367	13333	19467	128	17072	2.375
8	2	DQN	1203.933	0.970564	32339	461	549	16651	1.441
8	3	Tabular-Q	1197.662	0.970564	32339	461	549	16651	1.420
8	4	Static-MedRF	1201.634	0.983505	32283	517	58	17142	33.466
9	1	Tabular-Q	1195.749	0.971490	32359	441	537	16663	1.511
9	2	DQN	1199.750	0.971490	32359	441	537	16663	1.518
9	3	Static-MedRF	1205.626	0.984817	32315	485	44	17156	33.209
9	4	CFSM	1199.494	0.638539	13553	19247	106	17094	2.278
10	1	CFSM	1194.630	0.634338	13235	19565	123	17077	2.248
10	2	Tabular-Q	1197.355	0.970581	32347	453	556	16644	1.420
10	3	Static-MedRF	1206.329	0.984092	32310	490	64	17136	31.948
10	4	DQN	1203.465	0.970581	32347	453	556	16644	1.444

13 Full paired contrasts and diagnostic analysis

Table S9. Method minus Static-MedRF, ten trace-matched differences per contrast. Detection differences are percentage points, not percentages relative to the baseline. The intervals are ordinary, unadjusted Student-t 95% intervals.

Method	Outcome	Mean difference	95% CI
CFSM	Energy (J)	-9.9450	[-12.7729, -7.1171]
CFSM	F1 (pp)	-34.7424	[-34.8585, -34.6262]
CFSM	Recall (pp)	-0.3291	[-0.3766, -0.2816]
CFSM	FPR (pp)	57.6262	[57.3297, 57.9227]
CFSM	BA (pp)	-28.9776	[-29.1344, -28.8209]
CFSM	Mean batch (ms)	-30.2577	[-30.6798, -29.8356]
CFSM	Max. temp. (°C)	0.1461	[-0.1844, 0.4766]
Tabular-Q	Energy (J)	-13.0681	[-16.4000, -9.7362]
Tabular-Q	F1 (pp)	-1.3425	[-1.3926, -1.2924]
Tabular-Q	Recall (pp)	-2.9541	[-3.0710, -2.8371]
Tabular-Q	FPR (pp)	-0.1738	[-0.1964, -0.1512]
Tabular-Q	BA (pp)	-1.3901	[-1.4425, -1.3378]
Tabular-Q	Mean batch (ms)	-31.1478	[-31.5855, -30.7100]
Tabular-Q	Max. temp. (°C)	-0.1948	[-0.5693, 0.1797]
DQN	Energy (J)	-6.2270	[-10.8535, -1.6005]
DQN	F1 (pp)	-1.3425	[-1.3926, -1.2924]
DQN	Recall (pp)	-2.9541	[-3.0710, -2.8371]
DQN	FPR (pp)	-0.1738	[-0.1964, -0.1512]
DQN	BA (pp)	-1.3901	[-1.4425, -1.3378]
DQN	Mean batch (ms)	-31.1131	[-31.5350, -30.6911]
DQN	Max. temp. (°C)	0.1461	[-0.2230, 0.5152]

The arithmetic mean of blockwise energy-saving percentages is 1.082170% for Tabular-Q and 0.514887% for DQN. Dividing the difference of method grand means by the Static-MedRF grand mean instead yields approximately 1.083% and 0.516%. Neither convention is substituted silently for the other. Absolute paired effects are the primary quantities.

Table S10. Exploratory order-adjusted energy contrasts (J), using categorical block, position, and method terms and block-clustered uncertainty. This post-hoc analysis does not replace the primary paired analysis or remove unmeasured environmental effects.

Method	Adjusted difference	95% CI
CFSM	-9.542	[-13.795, -5.289]
Tabular-Q	-12.523	[-16.906, -8.139]
DQN	-6.243	[-11.940, -0.547]

The supplied CSVs also preserve leave-one-block-out effects and exact two-sided sign-flip diagnostics over ten paired differences. The latter rely on the sign-symmetry/exchangeability assumptions of that diagnostic. A Holm correction across the three energy tests is explicitly post-hoc; it is not represented as a preregistered correction across all reported endpoints. None of these procedures incorporates instrument calibration uncertainty.

14 Action occupancy and detector selection

Table S11. Detector occupancy (% of 5,000 formal windows per method), mean true runtime switches, and mean initialization transitions per run. Initialization is not counted as runtime adaptation.

Method	TinyDT	LightLR	MedRF	HeavyMLP	Runtime	Initial
Static-MedRF	0.00	0.00	100.00	0.00	0.0	1.0
CFSM	74.20	2.28	1.26	22.26	9.8	0.0
Tabular-Q	0.00	100.00	0.00	0.00	0.0	1.0
DQN	0.00	100.00	0.00	0.00	0.0	1.0

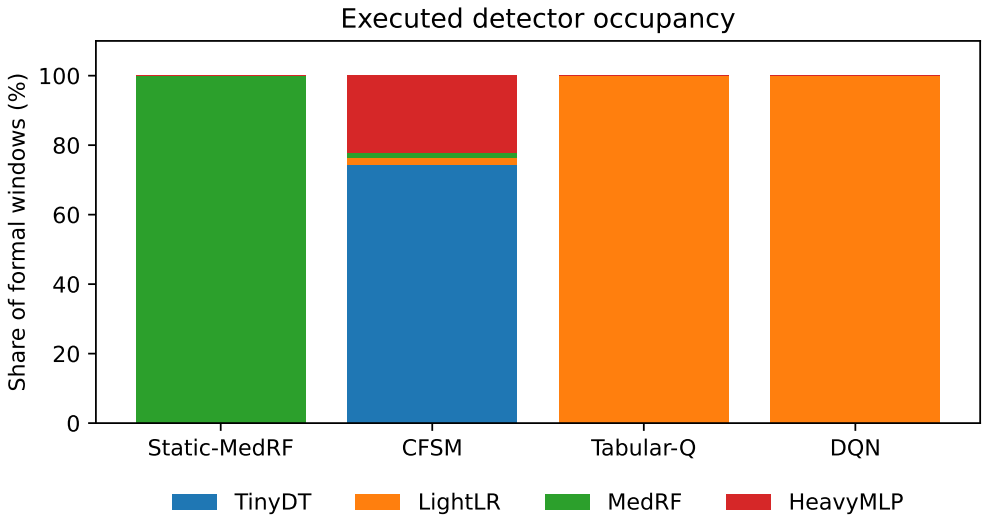


Figure S1. Both frozen learned policies execute a constant LightLR schedule. CFSM switches but spends most windows on the poorly performing TinyDT.

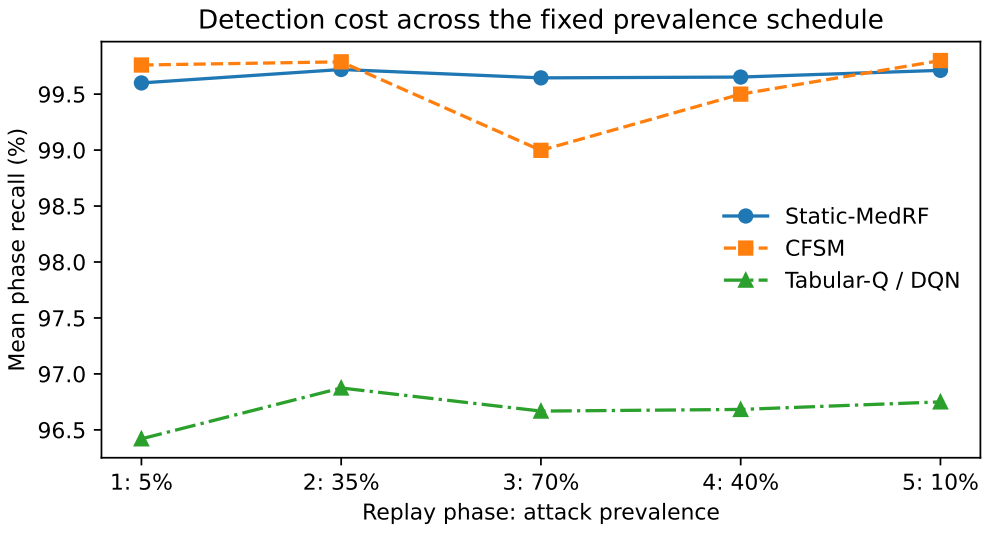


Figure S2. Mean recall by controlled prevalence phase. Tabular-Q and DQN produce identical selected predictions and are drawn as one series. These phases do not represent newly collected attacks or general concept drift.

15 Policy training and checkpoint selection

Tabular-Q and DQN each use one policy-training seed, 20260907, with 1,000 software episodes of 500 windows. The ten training traces and five validation traces are source-ID disjoint from the ten test traces. Validation is recorded every 25 episodes. The frozen Tabular-Q checkpoint comes from episode 50 and the DQN checkpoint from episode 25 (one-based numbering), not automatically from the final episode. Full curves, checkpoint metadata, and all 324 reconstructed greedy state actions are retained. The original harness did not instrument training energy. Archived process-start timestamps (06:10:58 and 06:11:41 UTC) and policy-metadata creation timestamps (06:11:24.687905 and 06:15:57.922987 UTC) indicate approximately 26.7 s for the Tabular-Q run and 256.9 s (4.28 min) for DQN. These are provenance-derived wall-clock intervals rather than calibrated compute benchmarks.

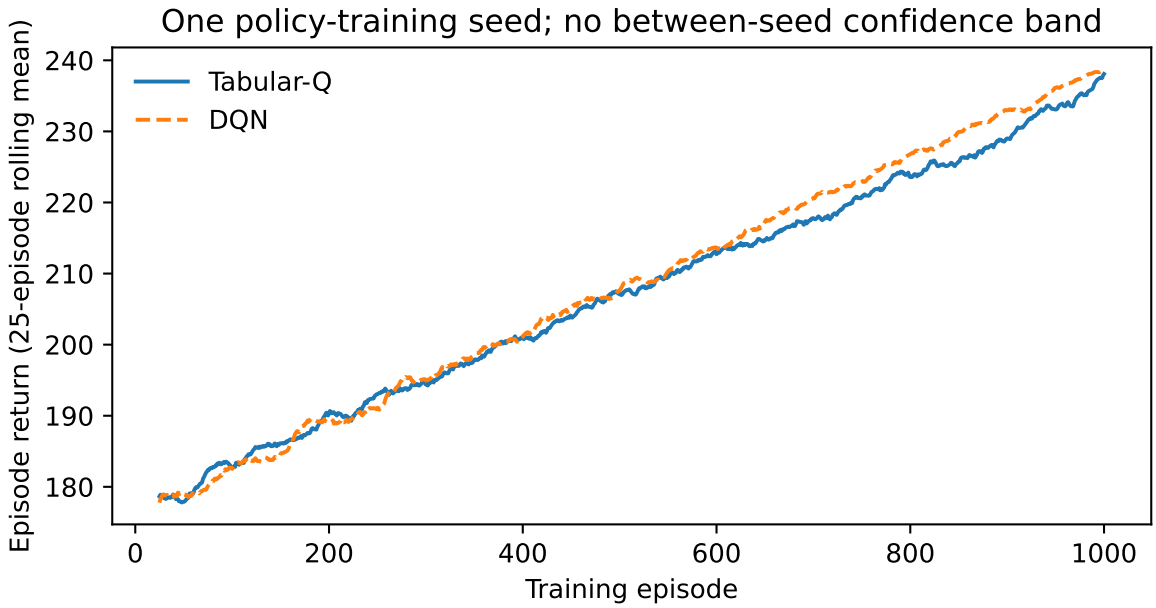


Figure S3. Training returns, displayed with a 25-episode trailing mean. Unsmoothed values are available in CSV. This is one realization per algorithm, not a multi-seed convergence guarantee.

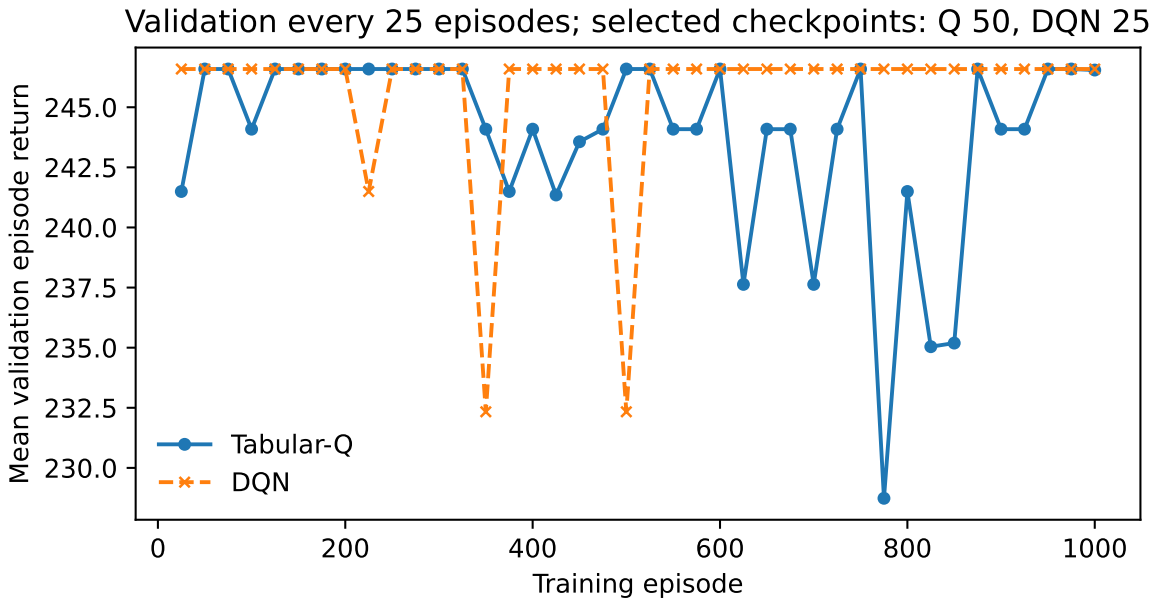


Figure S4. Recorded validation returns and checkpoint selection. The validation curve is not monotonic; the selected checkpoint is not evidence of convergence of every later iterate.

16 Post-hoc reward and state-coverage diagnostic

16.1 Pairwise non-degeneracy criterion

For two actions a, b, write the frozen scalar utility difference as

$$U_a(s) - U_b(s) = w_D \Delta D_{ab}(s) - \Delta C_{ab}, \qquad w_D > 0,$$

where ΔC_{ab} is the normalized non-detection resource-penalty difference on the declared support, and define $\tau_{ab} = \Delta C_{ab}/w_D$. Then

$$\operatorname{sign}(U_a(s) - U_b(s)) = \operatorname{sign}(\Delta D_{ab}(s) - \tau_{ab}).$$

Theorem S1 (Pairwise non-degeneracy)**.** *For $w_D > 0$, a strict preference reversal between actions a and b exists on support $\mathcal{S}$ if and only if*

$$\min_{s\in\mathcal{S}} \Delta D_{ab}(s) < \tau_{ab} < \max_{s\in\mathcal{S}} \Delta D_{ab}(s).$$

If this condition fails, every state in $\mathcal{S}$ gives the same sign of $U_a(s) - U_b(s)$.

Proof. Because $w_D > 0$, the sign of $U_a(s) - U_b(s)$ equals the sign of $\Delta D_{ab}(s) - \tau_{ab}$. A strict reversal requires states $s, s' \in \mathcal{S}$ with opposite signs, equivalently one detection difference below τ_{ab} and another above it. This is exactly the strict interval condition in Theorem S1. □

Corollary S2 (Bounded detection)**.** *If $D_a(s), D_b(s) \in [0, 1]$, reversal is impossible whenever $|\tau_{ab}| \geq 1$, or whenever the attainable interval $[\min \Delta D_{ab}, \max \Delta D_{ab}]$ does not straddle τ_{ab}.*

With multiple normalized resource terms, $\Delta C_{ab} = \sum_j w_j(\bar{C}_{j,a} - \bar{C}_{j,b})$. Candidate-pool min–max scaling therefore makes the implied security–resource exchange rate depend on pool membership; an external power, latency-SLO, energy, or thermal budget provides a stable deployment reference.

Algorithm S1 Validation-only reward audit

Require: Actions $\mathcal{M}$; declared reward weights; cost profile $\{C_a\}$; validation support $\mathcal{S}$; detection scores D_a
Ensure: Pairwise reversal-feasibility matrix R
1: **for** each unordered pair (a, b) **do**
2: $\tau_{ab} \leftarrow (C_a - C_b)/w_D$
3: $\Delta_{\min} \leftarrow \min_{s\in\mathcal{S}} \Delta D_{ab}(s)$; $\Delta_{\max} \leftarrow \max_{s\in\mathcal{S}} \Delta D_{ab}(s)$
4: $R_{ab} \leftarrow$ attainable iff $\Delta_{\min} < \tau_{ab} < \Delta_{\max}$
5: **end for**
6: Flag a reward as degenerate on $\mathcal{S}$ if no policy-relevant pair is attainable.

Algorithm S1 uses only quantities available at policy freeze and therefore does not inspect test outcomes. The supplied implementation verifies the configuration, manifest, and five validation-trace hashes, evaluates all four actions on each of 2,500 validation windows, and emits a pairwise CSV plus JSON audit. For MedRF minus LightLR it obtains $\tau = 0.79667561$ and an observed validation-window balanced-accuracy difference range of $[-0.10526316, 0.10526316]$, hence no strict reversal on that support. The separately reported aggregate test-set difference, 0.014011996, is explicitly tagged post-hoc descriptive and does not enter the audit. A failed audit calls for an audited static schedule or a pre-test redesign of the normalization, not post-hoc reward tuning.

Table S11A. Pairwise thresholds τ_{ab} under the frozen candidate-range normalization. Each value is the balanced-accuracy improvement action a must offer over b to overtake it.

$a\backslash b$	TinyDT	LightLR	MedRF	HeavyMLP
TinyDT	–	0.044	−0.753	−0.151
LightLR	−0.044	–	−0.797	−0.195
MedRF	0.753	0.797	–	0.602
HeavyMLP	0.151	0.195	−0.602	–

16.2 Executed reward decomposition

The fixed reward is $0.5\,\mathrm{BA} - 0.2\,\widetilde{P} - 0.2\,\widetilde{L} - 0.1\,\widetilde{H}$. The power min–max span is 0.049806200 W; the batch-latency span is 29.654800 ms. The compact software traces omit temperature telemetry, so the adapter supplies a constant 35 degrees Celsius predecision fallback and reuses that same, action-invariant value in the outcome field. The executed thermal penalty is consequently zero for every action; it is not a measured post-inference temperature rise and cannot provide action-level thermal attribution. The nonthermal resource penalty is 0.001662195 for LightLR and 0.4 for MedRF. Consequently, at equal thermal cost, MedRF requires an immediate balanced-accuracy improvement greater than 0.79667561 to exceed LightLR's single-window reward. This calculation explains the strength of the declared normalization; it is not a proof of a global optimal policy and did not lead to post-test reward changes.

Independent recomputation of the label-informed one-step reference selects LightLR in 4,991 of 5,000 held-out windows and HeavyMLP in nine, with neither MedRF nor TinyDT selected. This is a diagnostic reference using unavailable deployment labels, not a deployable comparison policy. Its CSV retains the objective and selected predictions.

All formal temperature and queue fields map to bin 0, and load maps to bin 1. Threat context varies, but there is no tested high-temperature or high-load region. The frozen DQN selects LightLR in every one of the 324 encoded states; the tabular policy selects LightLR in ten states and TinyDT in 314. Unvisited or zero-initialized tabular entries are not evidence of valid behavior outside the measured support. No thermal model was fitted or validated in this adaptive package, and no such performance is claimed.

17 Secondary pairwise and state-support diagnostics

The primary analysis compares each adaptive method with Static-MedRF because that contrast was fixed in the original forty-run matrix. The following pairwise calculations are secondary diagnostics derived from the same ten blocks. They are useful for interpreting the realized policy but are not substituted for the primary endpoint.

Table S12. Selected secondary paired contrasts from the original campaign. Differences are first method minus second method; intervals use ten block-level Student-t differences.

Contrast	Metric	Mean difference	95% CI
Tabular-Q − DQN	Energy (J)	-6.841	[-9.717, -3.965]
Tabular-Q − DQN	F1	0.000000	[0.000000, 0.000000]
Tabular-Q − DQN	Recall	0.000000	[0.000000, 0.000000]
Tabular-Q − DQN	FPR	0.000000	[0.000000, 0.000000]
Tabular-Q − CFSM	Energy (J)	-3.123	[-5.461, -0.785]
Tabular-Q − CFSM	F1	0.333998	[0.332680, 0.335317]
Tabular-Q − CFSM	Recall	-0.026250	[-0.027565, -0.024935]
Tabular-Q − CFSM	FPR	-0.578000	[-0.580961, -0.575039]

Tabular-Q and DQN have identical selected predictions, so their detection differences are exactly zero rather than statistically estimated near-zero quantities. Tabular-Q nevertheless uses 6.841 J less energy on average than DQN (95% CI 3.965–9.717 J in Tabular-Q's favor). Tabular-Q also uses 3.123 J less than CFSM (0.785–5.461 J) while producing much higher F1. These empirical mean relations do not identify isolated controller cost and do not generalize beyond the campaign.

Table S13. Greedy action map and observed state support for the two learned policies. "All" counts cover the 324 encoded states; "Observed" counts cover unique states reached during the formal learned-policy runs.

Method	All states	Observed	All TinyDT	All LightLR	Obs. TinyDT	Obs. LightLR
Tabular-Q	324	4	314	10	0	4
DQN	324	4	0	324	0	4

Tabular-Q and DQN each reach only four unique states in formal execution. DQN maps every encoded state to LightLR. Tabular-Q maps ten encoded states to LightLR and 314 to TinyDT, but all four states actually reached by the learned-policy runs are LightLR states. Because most tabular states are never visited and many Q entries retain initialization structure, the all-state tabular map is not treated as validated deployment behavior outside observed support.

The phase-level diagnostic reaches the same conclusion. Neither learned policy changes action at a replay-phase boundary. Relative to Static-MedRF, mean learned-policy recall is lower by approximately 2.85–3.18 percentage points across each of the five phases while FPR is lower by approximately 0.16–0.20 percentage points. Phase-varying aggregate metrics therefore describe a constant LightLR operating point under changing class prevalence, not learned phase adaptation.

18 Archive reconciliation and reproducibility limits

All 558 top-level adaptive manifest entries verify. The runtime manifest separately references 57 artifacts: 22 match supplied bytes directly, 22 trace index/label arrays are independently reconstructed byte-exactly, seven depend on the frozen external baseline, and six source paths originally differed from the later source copies packaged with the adaptive ZIP. A self-consistent final ZIP and a byte-identical execution snapshot are therefore distinct verification questions.

The six runtime paths are `controller.py`, `worker.py`, `scheduler/artifacts.py`, `scheduler/cli.py`, `scheduler/evaluation.py`, and `scheduler/trace_io.py`. An archival-only V2 recovery package now contains all six historically executed byte sequences. Independent verification confirms 6/6 exact SHA-256 matches to the runtime-bound expected values, verifies all 17 content entries in its manifest, and retains the later source versions separately. No experiment was rerun and no V1 evidence was modified for this closure.

The historical primary archive itself does not isolate scheduler decision overhead. That gap is addressed only by the separate P1B corrected microbenchmark described below; source recovery alone does not add measurement evidence.

19 P1A matched Static-LightLR physical control

P1A is a post-hoc attribution experiment designed after the primary learned policies were observed to select LightLR throughout. It contains ten matched blocks with Static-LightLR, Tabular-Q, DQN, and Static-MedRF. The archive retains 42 attempted runs: 40 valid formal runs, one DQN preparation failure before workload execution, and one Static-MedRF interruption after 403/500 power samples. The interrupted run is linked to exactly one authorized `retry1` replacement. No invalid attempt is silently deleted or combined with its replacement.

For every valid Tabular-Q, DQN, and Static-LightLR run, the selected-probability hash, confusion counts, and detection metrics are identical within the matched block. Each valid run contains 501 power observations and 500 decisions. Independent trapezoidal reintegration from the raw monotonic timestamps reproduces the registered run energy.

Table S14. P1A paired effects. Differences are first method minus second method. Exact sign-flip p is a secondary robustness diagnostic; Student-t intervals are the primary uncertainty summary.

Contrast	Outcome	Mean	95% CI	Sign-flip p
Tabular–Q – Static–LightLR	Energy (J)	0.4555	[-1.8609,2.7719]	0.6641
Tabular–Q – Static–LightLR	Power (mW)	0.9096	[-3.7224,5.5416]	0.6680
Tabular–Q – Static–LightLR	F1 (pp)	0.0000	[0.0000,0.0000]	1.0000
Tabular–Q – Static–LightLR	Recall (pp)	0.0000	[0.0000,0.0000]	1.0000
Tabular–Q – Static–LightLR	FPR (pp)	0.0000	[0.0000,0.0000]	1.0000
DQN – Static–LightLR	Energy (J)	4.5251	[2.6933,6.3569]	0.0020
DQN – Static–LightLR	Power (mW)	9.0452	[5.3812,12.7092]	0.0020
DQN – Static–LightLR	F1 (pp)	0.0000	[0.0000,0.0000]	1.0000
DQN – Static–LightLR	Recall (pp)	0.0000	[0.0000,0.0000]	1.0000
DQN – Static–LightLR	FPR (pp)	0.0000	[0.0000,0.0000]	1.0000
Static–MedRF – Static–LightLR	Energy (J)	12.4706	[10.6105,14.3306]	0.0020
Static–MedRF – Static–LightLR	Power (mW)	24.9412	[21.2208,28.6615]	0.0020
Static–MedRF – Static–LightLR	F1 (pp)	1.3425	[1.2924,1.3926]	0.0020
Static–MedRF – Static–LightLR	Recall (pp)	2.9541	[2.8371,3.0710]	0.0020
Static–MedRF – Static–LightLR	FPR (pp)	0.1738	[0.1512,0.1964]	0.0020
Tabular–Q – DQN	Energy (J)	-4.0696	[-6.4580,-1.6813]	0.0098
Tabular–Q – DQN	Power (mW)	-8.1356	[-12.9068,-3.3644]	0.0098
Tabular–Q – DQN	F1 (pp)	0.0000	[0.0000,0.0000]	1.0000
Tabular–Q – DQN	Recall (pp)	0.0000	[0.0000,0.0000]	1.0000
Tabular–Q – DQN	FPR (pp)	0.0000	[0.0000,0.0000]	1.0000

The Tabular-Q minus Static-LightLR energy contrast is +0.4555 J with interval $[-1.8609, 2.7719]$ J. Because no equivalence margin was predeclared, this is reported only as no resolved directional difference. DQN minus Static-LightLR is +4.5251 J with interval [2.6933, 6.3569] J. Static-MedRF minus Static-LightLR is +12.4706 J [10.6105, 14.3306] J and has higher detection quality. The supplementary campaign therefore shows that direct LightLR accounts for essentially the same detection operating point as both learned policies, while DQN carries an additional whole-device energy premium under the retained runtime.

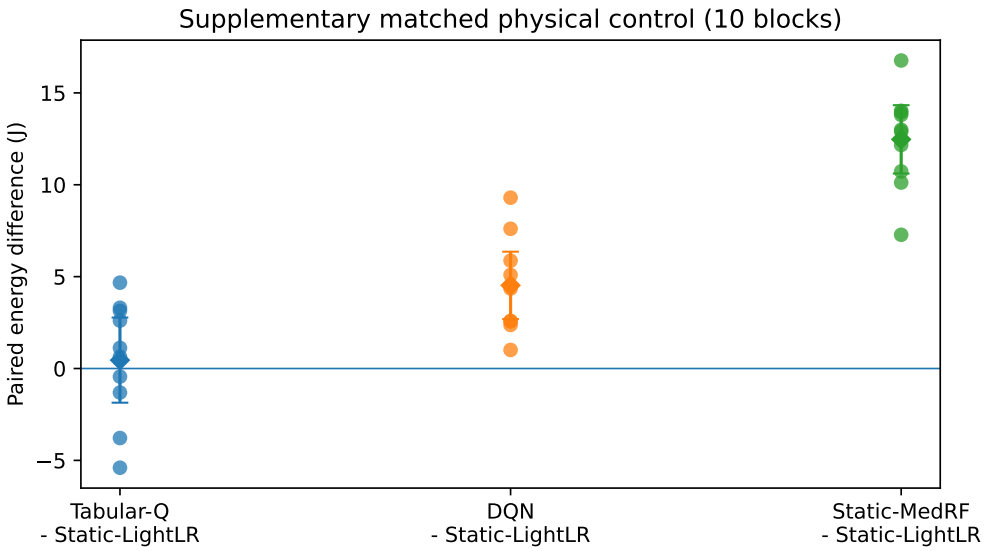


Figure S5. Ten P1A blockwise energy differences with Student-t 95% intervals.

20 P1B corrected isolated scheduler-overhead benchmark

The first P1B attempt contained 240 timing units but failed the final validator because its CFSM path substituted fixed MedRF behavior and the required telemetry was absent. The complete attempt remains in the archive and is excluded from inference. A single authorized corrected replacement repeats the frozen benchmark design, adds telemetry, and passes 240/240 units. A deterministic 5,000-state comparison between the corrected CFSM and the frozen rule reports zero action mismatches.

Each method is measured in 30 paired benchmark blocks on each of two state streams: all 324 encoded states and the state support observed in the held-out traces. Each block executes 1,000 warm-up decisions and 10,000 timed state-to-action decisions. The raw nanosecond arrays reproduce the registered mean, median, p95, p99, and sample standard deviation exactly. The repeat unit is the benchmark block, not the 10,000 individual timings.

Table S15. P1B corrected decision overhead. The CI is across 30 block means; p95/p99 are means of block-level quantiles.

Stream	Method	Mean (μs)	95% CI	p95 (μs)	p99 (μs)
exhaustive_324	CFSM	6.245	6.228–6.263	6.614	6.726
exhaustive_324	DQN	1559.645	1552.929–1566.362	1622.594	1651.166
exhaustive_324	Static-LightLR	0.634	0.626–0.641	0.689	0.842
exhaustive_324	Tabular-Q	10.443	10.420–10.466	10.643	10.947
observed_support	CFSM	6.582	6.566–6.598	6.942	7.172
observed_support	DQN	1566.897	1560.904–1572.890	1629.629	1668.873
observed_support	Static-LightLR	0.739	0.716–0.761	0.952	1.127
observed_support	Tabular-Q	10.796	10.764–10.828	11.047	11.286

On observed support the means are 0.739 μs for Static-LightLR, 6.582 μs for CFSM, 10.796 μs for Tabular-Q, and 1.5669 ms for DQN. DQN is about 145 times the Tabular-Q state-to-action time but remains approximately 0.157% of a one-second decision interval. These timings are not converted into energy.

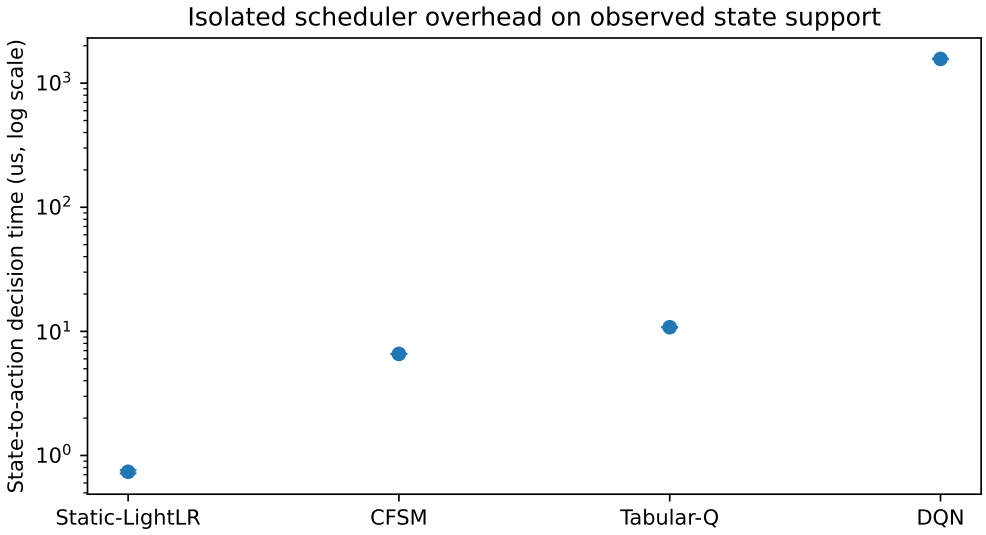


Figure S6. Corrected P1B state-to-action timing on observed state support. The vertical axis is logarithmic.

21 P1C multi-seed frozen-policy robustness

P1C trains ten additional Tabular-Q and ten additional DQN policies under the frozen state, reward, data split, training horizon, and validation-only checkpoint rule. During final closure, the missing derived validation/test registry is produced from those already frozen checkpoints without retraining or checkpoint reselection. All twenty policies select LightLR in 100% of held-out windows. Thus each algorithm has 10/10 held-out LightLR collapses, with Wilson 95% interval [0.7225, 1.0000]. Raw switch counts include one initialization event per held-out trace; between-window switches are zero.

This robustness is support-limited rather than global. The action maps over all 324 encoded states vary across seeds, particularly for DQN. Table S16 reports the complete map counts while the final column reports held-out LightLR occupancy.

Table S16. P1C greedy global action maps and held-out occupancy. Counts cover all 324 encoded states.

Algorithm	Seed	TinyDT	LightLR	MedRF	HeavyMLP	Held-out LR (%)
DQN	1009	0	324	0	0	100.0
DQN	1013	205	119	0	0	100.0
DQN	1019	0	324	0	0	100.0
DQN	1021	0	324	0	0	100.0
DQN	1031	85	239	0	0	100.0
DQN	1033	192	10	0	122	100.0
DQN	1039	0	324	0	0	100.0
DQN	1049	311	13	0	0	100.0
DQN	1051	67	139	0	118	100.0
DQN	1061	0	324	0	0	100.0
Tabular-Q	1009	314	10	0	0	100.0
Tabular-Q	1013	313	11	0	0	100.0
Tabular-Q	1019	313	11	0	0	100.0
Tabular-Q	1021	312	12	0	0	100.0
Tabular-Q	1031	312	12	0	0	100.0
Tabular-Q	1033	314	10	0	0	100.0
Tabular-Q	1039	312	12	0	0	100.0
Tabular-Q	1049	312	12	0	0	100.0
Tabular-Q	1051	313	11	0	0	100.0
Tabular-Q	1061	313	11	0	0	100.0

22 P1D reward/state sensitivity and P1E hyperparameter sensitivity

P1D contains 85 valid policies plus one retained interrupted DQN attempt. Every valid policy is single-action on held-out support. Of the 85, 80 select LightLR. The only exceptions are the five Tabular-Q `detection_only` reward policies, which select MedRF throughout and recover the MedRF-like F1. All thirty Tabular-Q state-ablation policies select LightLR. This supports specification sensitivity in a precise sense: the tested resource-aware objectives and state removals preserve the LightLR operating point, while deleting all resource penalties changes which static detector is preferred.

Table S17. P1D held-out policy results by family/variant. "LR" is the number of LightLR-collapsed policies; "Single" counts any single-action policy.

Family	Algorithm	Variant	n	LR	Single	Mean F1
Reward	DQN	default	5	5/5	5/5	0.970439
Reward	DQN	detection_heavy	5	5/5	5/5	0.970439
Reward	DQN	energy_heavy	5	5/5	5/5	0.970439
Reward	DQN	equal	5	5/5	5/5	0.970439
Reward	Tabular-Q	default	5	5/5	5/5	0.970439
Reward	Tabular-Q	detection_heavy	5	5/5	5/5	0.970439
Reward	Tabular-Q	detection_only	5	0/5	5/5	0.983864
Reward	Tabular-Q	energy_heavy	5	5/5	5/5	0.970439
Reward	Tabular-Q	equal	5	5/5	5/5	0.970439
Reward	Tabular-Q	latency_heavy	5	5/5	5/5	0.970439
Reward	Tabular-Q	thermal_heavy	5	5/5	5/5	0.970439
State	Tabular-Q	full	5	5/5	5/5	0.970439
State	Tabular-Q	minus_arrival	5	5/5	5/5	0.970439
State	Tabular-Q	minus_previous_action	5	5/5	5/5	0.970439
State	Tabular-Q	minus_queue	5	5/5	5/5	0.970439
State	Tabular-Q	minus_temperature	5	5/5	5/5	0.970439
State	Tabular-Q	minus_threat	5	5/5	5/5	0.970439

P1E contains 35 valid Tabular-Q runs over the predeclared discount/learning-rate matrix. All 35 select LightLR throughout held-out support and are single-action. These results are sensitivity evidence only; none is used to choose a replacement primary hyperparameter after test inspection.

Table S18. P1E Tabular-Q hyperparameter sensitivity on held-out support.

Variant	n	LR	Single	Mean F1
alpha_0.05	5	5/5	5/5	0.970439
alpha_0.10	5	5/5	5/5	0.970439
alpha_0.20	5	5/5	5/5	0.970439
gamma_0.00	5	5/5	5/5	0.970439
gamma_0.50	5	5/5	5/5	0.970439
gamma_0.90	5	5/5	5/5	0.970439
gamma_0.99	5	5/5	5/5	0.970439

23 R1 fixed-rate confidence cascade

R1 contains 40 valid 500-second runs in ten paired blocks. Static-LightLR, the frozen Tabular-Q policy, Static-MedRF, and the confidence cascade use the same trace within block; every method has an immediately preceding idle segment. The cascade band is fixed at $[0.3, 0.7]$ and ground truth is never used for escalation. It sends 1.3388% of rows to MedRF.

Table S19. R1 method means. Adjusted power subtracts the immediately preceding idle.

Method	Energy (J)	Power (W)	Adjust. (mW)	F1	Recall (%)	FPR (%)	Esc. (%)
Static-LightLR	1360.4	2.721	37.0	0.9704	96.71	1.36	–
Cascade	1368.3	2.737	50.6	0.9839	99.39	1.38	1.34
Tabular-Q	1362.0	2.724	39.7	0.9704	96.71	1.36	–
Static-MedRF	1368.6	2.737	56.5	0.9839	99.66	1.54	–

Against Static-LightLR, paired cascade effects are +7.952 J [5.897,10.007], +15.917 raw mW [11.798,20.035], and +13.635 idle-adjusted mW $[-1.515, 28.784]$. F1/recall/FPR differences are +1.347/+2.676/+0.018 percentage points. The raw and adjusted results are both retained because the adjusted interval indicates sensitivity to ambient/carryover drift.

The frozen worker implements sub-batch escalation, not one MedRF call per deferred row: it evaluates LightLR on the full 100-row batch and invokes MedRF once on the subset satisfying $0.3 \le p_{LR} \le 0.7$. The apparently disproportionate latency is explained by call granularity. Only 1.3388% of rows are deferred, but they occur in 66.76% of windows; among those windows the mean deferred subset has only 2.01 rows. A no-defer window averages 1.636 ms, whereas any-defer windows average 31.884 ms, nearly independent of whether one or several rows are deferred. Thus the 21.830-ms overall mean reflects frequent small MedRF calls at this 100-row cadence rather than row-by-row execution. It is an implementation/operating-point result, not a general bound for optimized cascades.

Table S19a. R1 confidence-cascade call-granularity diagnostic reconstructed from 5,000 frozen window logs.

Quantity	Value
Escalated rows / all rows	6,694 / 500,000 (1.3388%)
Windows with ≥ 1 escalated row	3,338 / 5,000 (66.76%)
Mean escalated rows per active window	2.01
No-escalation window inference	1.636 ms
Escalation-active window inference	31.884 ms
All cascade windows, mean inference	21.830 ms

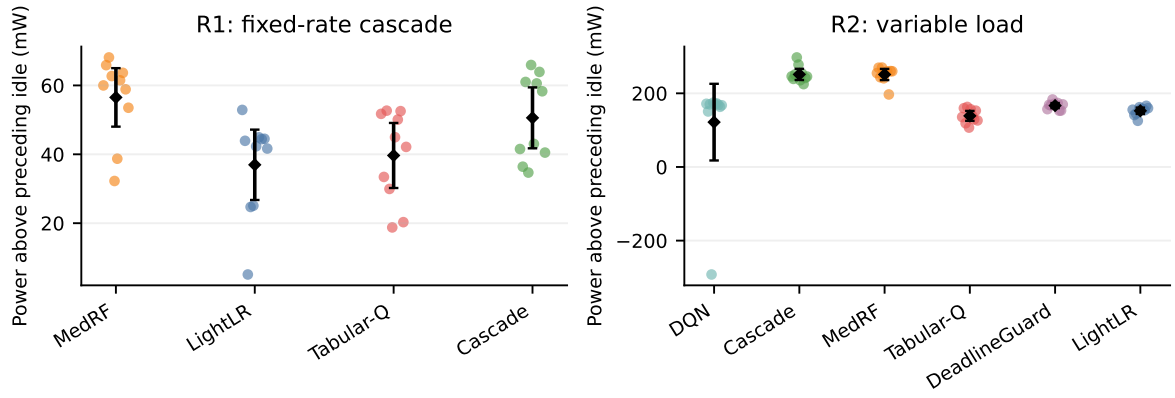


Figure S7. R1/R2 preceding-idle-adjusted power. Small points are runs; diamonds and bars are method means and Student-t 95% intervals over runs. Pairwise inference uses within-block differences in the CSV, not these marginal intervals.

24 R2 variable-load factorial and DeadlineGuard

Each of the 60 valid runs visits every 50/100/4000 rows/s × 0.10/0.50/0.90 prevalence cell exactly once, then the fixed recovery cell. The controller sees causal measurements rather than cell identity. Static-MedRF misses 512 of 5,000 deadlines and Cascade misses 21; DeadlineGuard and the remaining methods miss none. All final backlogs are zero, so deadline misses and end-of-run backlog are distinct endpoints.

Table S20. R2 method results over ten blocks. Energy is mean±SD across physical runs; completion is seconds.

Method	Energy (J)	Power (W)	F1	Recall (%)	FPR (%)	Miss (%)	p95 (s)
Static-LightLR	1416.8±3.7	2.834	0.9763	96.68	1.35	0.00	0.070
Cascade	1448.1±46.4	2.896	0.9902	99.41	1.36	0.42	0.665
DeadlineGuard	1424.4±5.2	2.849	0.9769	96.80	1.36	0.00	0.069
Tabular-Q	1400.9±48.0	2.802	0.7434	99.76	67.76	0.00	0.063
DQN	1461.1±123.9	2.922	0.9763	96.68	1.35	0.00	0.072
Static-MedRF	1468.1±4.8	2.936	0.9906	99.65	1.52	10.24	1.013

Table S21. R2 paired effects against Static-LightLR. Energy and adjusted-power intervals use ten within-block differences. F1 is percentage points; p is the exact two-sided sign-flip diagnostic for energy.

Method	ΔE (J), 95% CI	ΔP_{adj} (mW), 95% CI	ΔF1	p
Cascade	+31.3 [-1.4,+63.9]	+98.8 [+77.9,+119.8]	+1.38	0.0762
DeadlineGuard	+7.6 [+5.0,+10.1]	+14.2 [+0.4,+27.9]	+0.06	0.0020
Tabular-Q	-15.9 [-49.9,+18.1]	-13.8 [-32.7,+5.1]	-23.29	0.2637
DQN	+44.2 [-43.2,+131.7]	-30.8 [-128.1,+66.5]	+0.00	0.0020
Static-MedRF	+51.3 [+49.5,+53.0]	+98.6 [+79.3,+117.9]	+1.43	0.0020

Table S22. R2 state/action and service evidence. Occupancy totals 5,000 windows/method. Switch counts include the implementation's initialization convention.

Method	States	Action occupancy	Switches	Fallback	Misses	Backlog
Static-LightLR	15.4	LightLR:5000	1.0	0	0	0
Cascade	15.4	ConfidenceCascade_0p3_0p7:5000	1.0	0	21	0
DeadlineGuard	15.4	LightLR:1500, MedRF:3500	5.3	0	0	0
Tabular-Q	10.7	TinyDT:5000	0.0	5000	0	0
DQN	15.6	LightLR:5000	1.0	0	0	0
Static-MedRF	15.2	MedRF:5000	1.0	0	512	0

DeadlineGuard selects MedRF in 3,500 and LightLR in 1,500 windows and records 5.3 switches/run. Against Static-MedRF it uses 43.714 J less per run [41.418,46.010], reduces deadline misses by 10.24 percentage points to zero, and loses 1.373 F1 pp. Frozen DQN remains LightLR. Frozen Tabular-Q selects TinyDT in all 5,000 windows. The registry calls these selections "fallback", but the frozen source contains no separate exception branch: unseen zero-initialized Q rows are tied, NumPy argmax chooses the first maximum, and the action order places TinyDT first. The R2 behavior is therefore the direct hardware realization of the off-support tie/default already visible in the 314/324 TinyDT full greedy map.

DQN's raw energy difference from Static-LightLR is positive in all ten blocks, with median +5.72 J, but Block 4 is +392.0 J and raises the SD of paired differences to 122.2 J; the Student-t 95% interval consequently spans $[-43.2, 131.7]$ J. The same run has a 3.919-W preceding idle and 49.2°C maximum temperature. Preceding-idle-adjusted power does not resolve a DQN–LightLR difference. We therefore retain both the 10/10 raw direction (exact sign-flip $p = 0.00195$) and the unstable magnitude rather than collapsing them into an "equivalent" or causal controller-power claim.

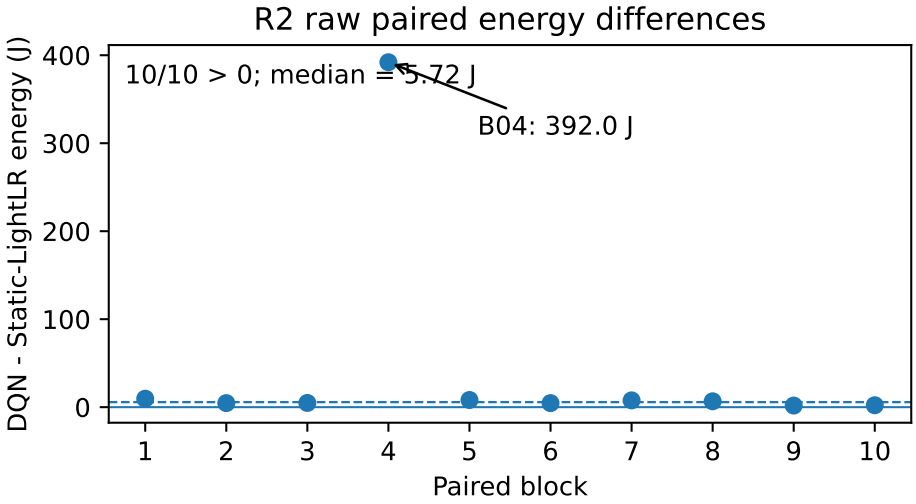


Figure S8a. R2 DQN-minus-Static-LightLR raw energy by paired block. Nine differences are between 1.8 and 9.6 J; Block 4 is a 392.0-J power/temperature excursion.

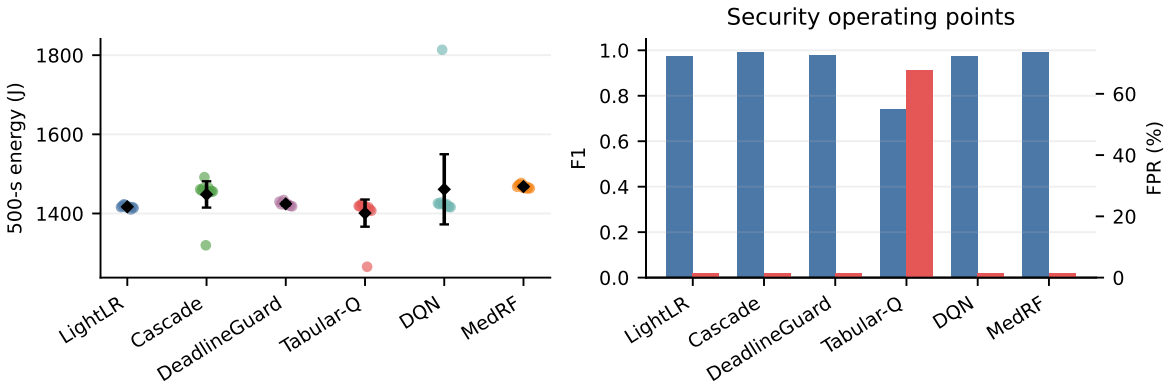


Figure S8. R2 raw energy distributions and security operating points. The block is the repeat unit.

25 R3 stop and R4 TinyDT threshold diagnostic

R3 required four original observed-support states for controller-only whole-device measurement. The frozen source exposes eight and contains no outcome-independent rule for selecting four. The campaign therefore stopped before measurement. No controller-only power or joule value is inferred from P1B timing.

R4A selects threshold 0.6467505097 solely on validation balanced accuracy with the highest-threshold tie rule, freezes its hash, and applies it once to test. R4B is not executed because an independent stable reference load is unavailable.

Table S23. TinyDT test operating point at the primary and diagnostic thresholds. The diagnostic does not replace the primary.

Result	τ	TN	FP	FN	TP	F1	Rec.	FPR	BA
Primary fixed	0.500	5519	11610	13	5348	0.4792	99.76	67.78	0.6599
Validation-selected diagnostic	0.646751	16817	312	21	5340	0.9698	99.61	1.82	0.9889

The diagnostic changes test F1 from 0.4792 to 0.9698 and FPR from 67.78% to 1.82% while recall changes from 99.76% to 99.61%. Test AUROC is 0.9906. This supports calibration/threshold sensitivity as an explanation for much of the threshold-0.5 failure, but the analysis was introduced after the discrepancy was known and cannot repair frozen policy outcomes.

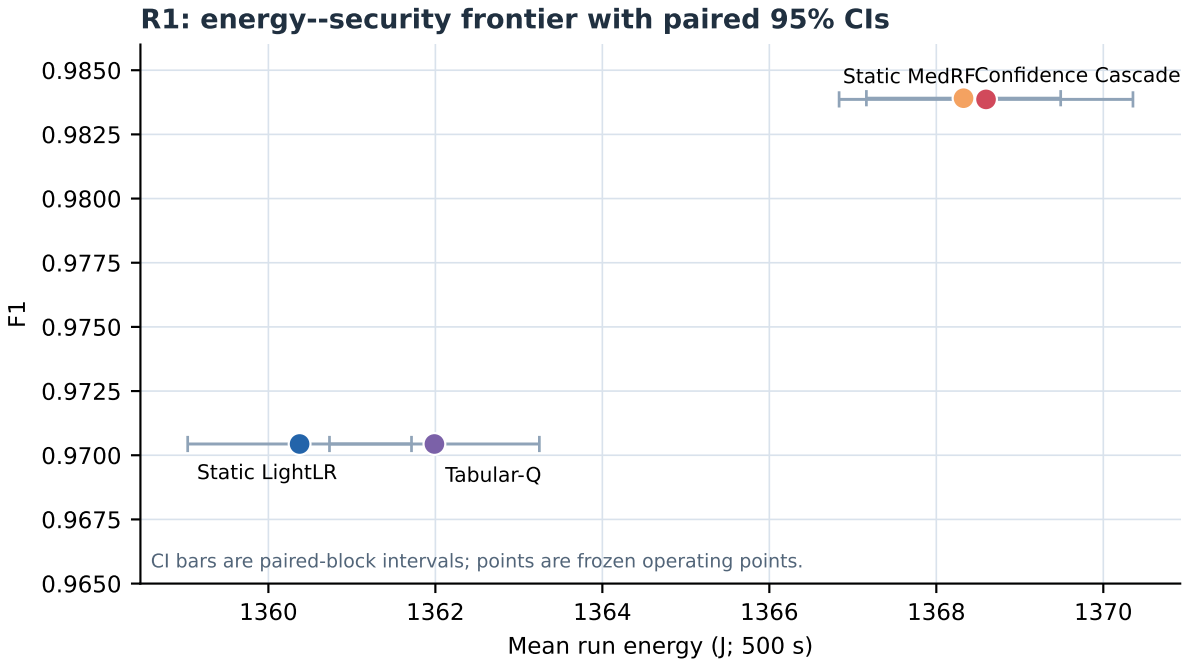


Figure S9. R1 energy–security frontier. Horizontal bars are paired-block 95% CIs for run energy; the plotted F1 values are method means.

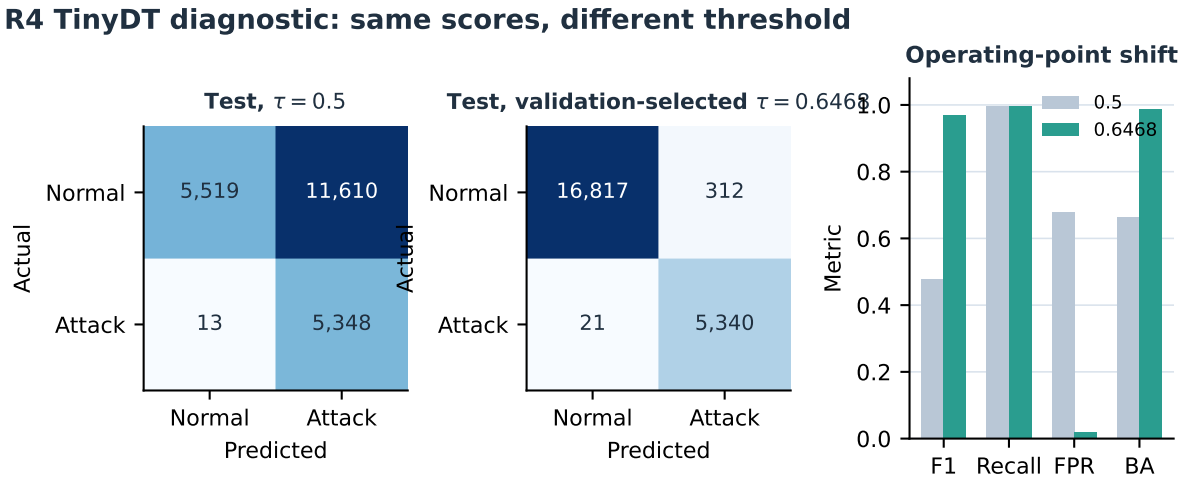


Figure S10. R4 TinyDT threshold diagnostic. The primary threshold remains 0.5; the validation-selected threshold is a post-hoc diagnostic only.

26 Recent literature context (2024–2026)

The main text now cites twenty recent peer-reviewed studies from CCF-A or CCF-B venues. The entries are deliberately used as contextual comparators rather than as evidence for this experiment's physical claims. Their common themes are edge orchestration, energy/deadline trade-offs, learning-based control, and robust offloading/caching. Venue classification follows the CCF 2022 journal catalogue (TMC=A; TSC=A); publication metadata and DOI links are retained in `references.bib`. None of these studies changes the frozen detector artifacts, split, thresholds, or physical measurements reported here.

Table S24. Added 2024–2026 literature set. All entries are IEEE Transactions on Mobile Computing (CCF-A) except Zhang *et al.*, IEEE Transactions on Services Computing (CCF-A).

Topic	Years	Keys
DVFS, startup-aware scheduling, distributed offloading	2024	[1, 2, 3]
Robust caching, dependency-aware and incentive-aware offloading	2024	[4, 5, 6, 7]
QoS, trust, D2D and random-access edge control	2024–25	[8, 9, 10, 11]
Energy harvesting, satellite/UAV/vehicular coordination	2024–25	[12, 13, 14, 15]
Digital-twin inference, distributed coding, uncertain inference, and edge caching	2024–26	[16, 17, 18, 19, 20]

27 Final package integrity

The final P0/P1 strengthening ZIP has SHA-256:

`d4e1e96960f1279c79445c744a02be1033e288dad27f440b87753934a3f4b108`

Its size is 141,322,551 bytes and it contains 1,918 ZIP entries. Its root manifest contains 1,917 entries and independently covers every content file except itself; CRC, manifest, nested-manifest, and final validator checks pass. The reviewer-revision evidence ZIP has SHA-256:

`845ced5c96f932ee8d3509e2effcb39870a2fb9995bad9a6fe524f6c60b270e4`

It has 1,912 entries and a 1,911-file root manifest. Its final validator reports R0/R1/R2/R4 PASS, R3 as a predeclared-ambiguity stop, no policy retraining, no test-driven selection, and retained invalid/replacement linkage. These properties establish internal provenance, not external validity.

References

[1] Z. Zhang, Y. Zhao, H. Li, C. Lin, and J. Liu, "Dvfo: Learning-based dvfs for energy-efficient edge-cloud collaborative inference," *IEEE Transactions on Mobile Computing*, vol. 23, no. 10, pp. 9042–9059, 2024.

[2] J. Lou, Z. Tang, W. Jia, W. Zhao, and J. Li, "Startup-aware dependent task scheduling with bandwidth constraints in edge computing," *IEEE Transactions on Mobile Computing*, vol. 23, no. 2, pp. 1586–1600, 2024.

[3] M. Kim, J. Jang, Y. Choi, and H. J. Yang, "Distributed task offloading and resource allocation for latency minimization in mobile edge computing networks," *IEEE Transactions on Mobile Computing*, vol. 23, no. 12, pp. 15 149–15 166, 2024.

[4] Y. Chen, M. Liu, B. Ai, Y. Wang, and S. Sun, "Adaptive bitrate video caching in uav-assisted mec networks based on distributionally robust optimization," *IEEE Transactions on Mobile Computing*, vol. 23, no. 5, pp. 5245–5259, 2024.

[5] C. Feng, P. Han, X. Zhang, Q. Zhang, Y. Liu, and L. Guo, "Dependency-aware task reconfiguration and offloading in multi-access edge cloud networks," *IEEE Transactions on Mobile Computing*, vol. 23, no. 10, pp. 9271–9288, 2024.

[6] F. Tutuncuoglu and G. Dan, "Joint resource management and pricing for task offloading in serverless edge computing," *IEEE Transactions on Mobile Computing*, vol. 23, no. 6, pp. 7438–7452, 2024.

[7] Y. Hu *et al.*, "Bridging incentives and dependencies: An iterative combinatorial auction approach to dependency-aware offloading in mobile edge computing," *IEEE Transactions on Mobile Computing*, vol. 23, no. 12, pp. 12 113–12 130, 2024.

[8] E. F. Maleki, W. Ma, L. Mashayekhy, and H. J. L. Roche, "Qos-aware content delivery in 5g-enabled edge computing: Learning-based approaches," *IEEE Transactions on Mobile Computing*, vol. 23, no. 10, pp. 9324–9336, 2024.

[9] J. Wang, Z. Li, H. Liu, T. Qiu, and H. Luo, "A trust-based computation offloading framework in mobile cloud-edge computing networks," *IEEE Transactions on Mobile Computing*, vol. 24, no. 6, pp. 5370–5385, 2025.

[10] Z. Hu, J. Niu, T. Ren, X. Liu, and M. Guizani, "Sitoff: Enabling size-insensitive task offloading in d2d-assisted mobile edge computing," *IEEE Transactions on Mobile Computing*, vol. 24, no. 3, pp. 1567–1584, 2025.

[11] R. Han, Y. Yu, Q. Zeng, J. Wang, L. Bai, J. Choi, and W. Zhang, "Offloading game for mobile edge computing with random access in iot," *IEEE Transactions on Mobile Computing*, vol. 24, no. 4, pp. 3316–3329, 2025.

[12] X. Zhang, J. Liu, R. Zhang, Y. Huang, J. Tong, N. Xin, L. Liu, and Z. Xiong, "Energy-efficient computation peer offloading in satellite edge computing networks," *IEEE Transactions on Mobile Computing*, vol. 23, no. 4, pp. 3077–3091, 2024.

[13] Y. Luo, L. Pu, and C.-H. Liu, "Computing power and battery charging management for solar energy powered edge computing," *IEEE Transactions on Mobile Computing*, vol. 24, no. 3, pp. 1913–1927, 2025.

[14] H. Gong, B. Huang, B. Jia, L. Hao, and Z. Shi, "Jointly optimizing the energy and time for multi-uav 3-d coverage of terrestrial regions," *IEEE Transactions on Mobile Computing*, vol. 24, no. 10, pp. 10 312–10 329, 2025.

[15] L. Zhao, L. Sun, A. Hawbani, Z. Liu, X. Tang, and L. Xu, "Dual dependency-aware collaborative service caching and task offloading in vehicular edge computing," *IEEE Transactions on Mobile Computing*, vol. 24, no. 10, pp. 10 963–10 977, 2025.

[16] Y. Zhang, W. Liang, Z. Xu, W. Xu, and M. Chen, "Aoi-aware inference services in edge computing via digital twin network slicing," *IEEE Transactions on Services Computing*, vol. 17, no. 6, pp. 3154–3170, 2024.

[17] H. Qiu, K. Zhu, D. Niyato, N. C. Luong, C. Yi, and C. Dai, "Barycentric coded distributed computing with flexible recovery threshold for collaborative mobile edge computing," *IEEE Transactions on Mobile Computing*, vol. 25, no. 4, pp. 5557–5570, 2026.

[18] Z. Nan, Y. Han, S. Zhou, and Z. Niu, "Robust dnn partitioning and resource allocation under uncertain inference time," *IEEE Transactions on Mobile Computing*, vol. 25, no. 3, pp. 3680–3696, 2026.

[19] R. Luo, Z. Zhang, Q. He, M. Xu, F. Chen, X. Dai, S. Wu, and H. Jin, "Cost-effective edge data caching with failure tolerance and popularity awareness," *IEEE Transactions on Mobile Computing*, vol. 24, no. 6, pp. 5357–5369, 2025.

[20] P. Li, H. Zhang, Y. Wu, L. Qian, R. Yu, D. Niyato, and X. Shen, "Filling the missing: Exploring generative ai for enhanced federated learning over heterogeneous mobile edge devices," *IEEE Transactions on Mobile Computing*, vol. 23, no. 10, pp. 10 001–10 015, 2024.